

Systematic Trading Lab

From a first Python script to a statistically validated intraday strategy, a research lab and a production trading bot on MetaTrader 5. Built by Ye alongside online quantitative-finance coursework.

I set out to build an automated trading system that could pass a prop-firm evaluation. Over ten months it grew from a paper-trading bot into a full research and execution stack: a data layer over three market-data sources, a broker-realistic backtester, a statistics engine that refuses to call luck an edge, a strategy search over 217 instruments, a prop-firm economics simulator, and a live bot whose decisions reproduce the backtest trade for trade. Most ideas failed along the way; the process for finding out *which* ideas fail is the main thing I built.

217

instruments searched
across 7 asset classes and
3 data sources

10,906

strategy configurations
from 14 published strategy
families

1 of 490

finalist strategies survived
every out-of-sample test
(one instrument, two rules)

2,080 / 2,080

backtest trades reproduced
exactly by the live bot's
own decision code

Timeline

WINTER 2025 – 26

First HFT and scalping tick bots

The first bots were scalpers: a Bollinger lower-band touch confirmed by Stochastic RSI below 20, traded only above the 200-period EMA, with +0.25% / -0.60% exits, a 0.02% slippage filter and up to 10 positions. Risk limits were in from the first version (pause at -3% on the day, stop at -10% drawdown) as it grew from 4 markets on 60-second scans to 22 markets on 30-second scans.

Python REST APIs Bollinger / Stoch RSI risk limits

EARLY 2026

Micro-dip tick scalper v1 – v2

Rebuilt for speed: an asyncio / aiohttp loop scanning every 0.5 seconds (20× faster than the 10-second bot before it), one batched quote request for all 22 markets per scan, buying 0.1% micro-dips for a +0.25% target. v2 added per-symbol cool-downs after a stop and a minimum-dollar-profit exit, and filled its first orders within two seconds of the opening bell.

asyncio / aiohttp batched quotes 0.5 s loop uv

What carried forward: a non-blocking event loop and hard risk limits set before the first trade.

EARLY SUMMER 2026

Broker APIs: eToro, then Alpaca

Started with the eToro API (Open API and the Agent Portfolios beta) - hit authentication and access limits (HTTP 403, read-only scopes), so moved execution to Alpaca's paper-trading API.

eToro API REST auth Alpaca API Python

SUMMER 2026

Paper-trading bots v1 – v6

Async Python bots (aiohttp) scanning 20+ stocks and crypto every few seconds. The defining bug: a P&L counter showing six figures while real account equity hadn't moved - the bot treated *submitted* orders as *filled* positions. Rebuilt order handling around confirmed fills: fill confirmation with timeout, stale-order cancellation, startup position recovery, partial fills, loss-streak circuit breakers, market-hours from the broker's clock. v6 added a 1-10 confidence-scoring engine and shorting.

asyncio / aiohttp order state machine VS Code PowerShell

Lesson: the broker's account is the only source of truth - never a private ledger.

SUMMER 2026

TradingView, Pine Script and webhooks

Wrote Pine Script strategies (VWAP + regime detection, a multi-signal strategy) and a "market context" indicator that sends JSON alerts through webhooks to a Flask receiver (ngrok tunnel) that executes on Alpaca. The plumbing worked; the backtests were negative.

Pine Script v5 TradingView alerts Flask ngrok Claude Code

Lesson: execution infrastructure doesn't create an edge. Prove the signal first.

AUGUST 2026

A rules-based momentum bot and the first serious backtests

A three-rule 5-minute momentum system (trend, momentum band, volume) with 2-bar confirmation and a liquidity-sweep filter. Production hardening: earnings guard, slippage guard, live position monitor, Windows keep-awake, and a timezone bug that silently blocked trades. Market data moved from tvdatafeed to Yahoo Finance; 2- and 6-year backtests across 18–29 tickers; a 10,000-path Monte Carlo to size a portfolio against prop-firm drawdown limits.

yfinance pandas / numpy Monte Carlo Alpaca paper (\$25k)

SEPTEMBER 2026

Into MetaTrader 5 and prop firms

To trade prop-firm capital the bot had to run on MT5. First version: an MQL5 (C++-based) Expert Advisor acting as a file bridge - publishing live 5-minute bars to CSV and executing order/close requests from Python. Then the native MetaTrader5 Python API. Traced a terminal that wouldn't connect to a corrupted profile. Evaluated execution routes (TradersPost, SignalStack) and an optimizer across 67 instruments with out-of-sample portfolio construction.

MQL5 (C++) MetaTrader5 Python API FTMO prop-firm rules

LATE SEPTEMBER 2026

Production execution bot + pre-registered experiments

A live loop built to not lose money by accident: server-clock measurement (the broker's clock moves with New York DST), closed-bar logic, lot sizing asked from MT5 itself, stops on the broker from the first moment, prop-firm loss floors recomputed from the broker's numbers every loop, a worst-case "every stop hits" check, a kill switch, a high-impact news blackout and a 44-test suite against a fake MT5. Two new strategy ideas were tested against pass bars fixed in advance (23,051 and 148,197 simulated trades); both failed and were dropped, not rescued.

pytest fake-broker testing ForexFactory feed pre-registration

5 OCTOBER 2026

The audit that stopped a challenge purchase

Re-checking the backtester against how the broker really behaves: stops had only been checked inside the trading session and gap losses were capped at -1R, while the live bot's stops sit on the server 24 hours a day. Re-simulated with round-the-clock stops and real gap fills, the strategy's edge disappeared. Decision: don't buy a challenge on it.

Lesson: a backtest is only as honest as its fill model. The simulator must match the broker.

OCTOBER 2026

The research lab

One tool for data, backtests and verdicts. Data from MT5, Alpaca (SIP minute bars since 2016) and Yahoo, cached and cross-checked. A broker-realistic engine: next-bar fills, bid/ask, per-bar spreads, swaps, gap fills, 1-minute exits. Verdicts from fixed rules: block-bootstrap intervals, deflated Sharpe, a random-entry test (does it beat random entries in the same month and hour?), a robustness battery, an automatic look-ahead check and a 12-month vault. Its self-test: luck called an edge 2–4% of the time; a planted real edge found 87–93% of the time.

Then a strategy search: 14 strategy families from published research × 217 instruments × 10,906 configurations. Discovery on pre-2022 data only → 490 finalists → confirmation 2022–2025 with false-discovery control → an independent price feed → the held-back final year. One intraday momentum effect on a single US stock passed everything; the same rules on 46 other stocks did not.

numba block bootstrap Benjamini–Hochberg deflated Sharpe pyarrow

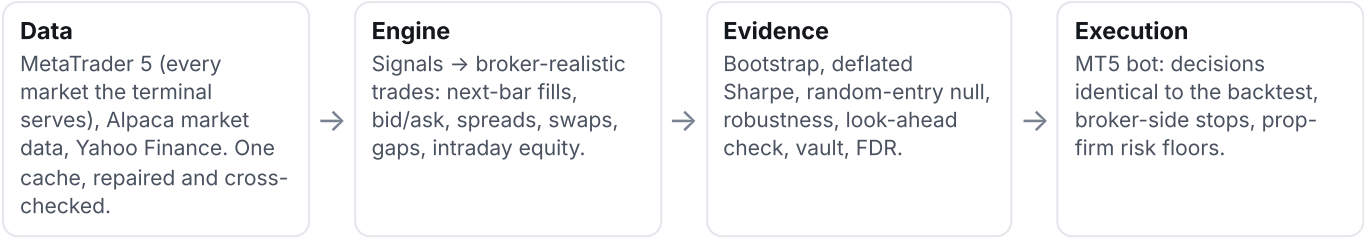
OCTOBER 2026

From edge to account

A prop-firm lifecycle simulator (fees, phases, payouts, splits, scaling, breaches and re-buys) across 11 programs, plus which firms allow automation and offer the instrument. Finding: holding risk fixed beats cutting it in drawdowns once fees are counted. A new execution bot was built around the validated rules; replaying its decision code over the history reproduces all 2,080 backtest trades exactly. Next: an FTMO trial account, then the first evaluation.

MT5 execution parity testing 26 tests

Architecture



Around it: a strategy search (discovery → confirmation → final year), a prop-firm economics simulator, and an HTML report per run. Everything is plain Python - no paid platforms.

Strategies tested end-to-end (top six)

Strategy	Idea (published research)	Out-of-sample result	Status
Intraday momentum	Two breakout rules (volatility breakout, noise-area); long & short; flat at the close	Sharpe 1.3–1.7 per rule (2022–25), 1.1–1.6 on a second data feed, positive in the held-back year; beats random entries ($p \leq 0.002$)	Live
RSI(2) reversion	Connors' 2-day RSI washouts with a 200-day trend filter	Sharpe ~0.9, PF ~2.2, but only 74 trades	Needs data
Opening-range breakout	Trade the first break of the first N minutes' range	Sharpe 0.5–0.6; didn't beat random entries	Watch list
First-candle momentum	Follow the first 5-minute candle (Zarattini & Aziz)	Sharpe ~0.9 - but random entries did the same: market drift	Rejected
IBS reversion	Buy closes near the day's low	Positive 2022–25; negative on the second feed and the final year	Rejected
Gap follow-through	Large opening gaps continue	Positive 2022–25; failed the second feed and the final year	Rejected

Instruments, parameter values, sizing and prop-firm settings are deliberately not published.

Tech stack & connectors

Languages	Python 3.12, pandas, Pine Script v5, MQL5 (C++-based), PowerShell
Market data	MetaTrader 5, Alpaca Market Data (SIP), Yahoo Finance (yfinance), TradingView (tvdatafeed), ForexFactory calendar
Execution	MetaTrader5 Python API, MQL5 file-bridge EA, Alpaca Trading API, TradingView webhooks → Flask/ngrok, eToro API
Research	pandas, numpy, numba, pyarrow, bootstrap & multiple-testing statistics, Monte Carlo
Engineering	VS Code, Git/GitHub, pytest with a fake broker; vibe-coded with Claude Code, with an agentic AI workflow built in (e.g. pipeline discovery)

What's next

Forward-test the live bot on an FTMO trial and compare every fill with the backtest; then the first evaluation. Research: a basket RSI(2) test and monthly re-runs on new data.

Principles I work by now

- **Decide the pass bar before looking.** Verdict thresholds are fixed in code before a test runs.
- **Count every attempt.** Every variant tried raises the bar for the next (deflated Sharpe, FDR).
- **Beat random, not zero.** A strategy must beat random entries in the same market, month and hour.
- **Lock away a year.** The newest 12 months are opened once, at the end.
- **The fill model is the strategy.** Next-bar fills, real spreads, gaps through stops, 24-hour stops.
- **Live must equal backtest.** Parity tests replay the bot's decisions against the backtest.
- **Failures are results.** Three of my own strategies were dropped by these rules - that's the system working.

Figures are backtest and simulation results - not live trading results and not investment advice. The research lab and execution bot are on GitHub with strategy parameters removed.

The numbers behind the two live models

Broker's own CFD prices (bid/ask, per-bar spreads, commissions), Jan 2022 – Oct 2026: 1,241 trading days, all after the discovery period. 1R = the amount risked on one trade. Daily P&L in R; Sharpe annualised from daily returns ($\sqrt{252}$). Bracketed figures: an independent exchange price feed.

Metric	VB breakout	NA noise band	Combined
Trades (per year)	786 (160)	1,135 (230)	1,921 (390)
Win rate · payoff ratio	47.6% · 1.48	36.2% · 2.33	40.9% · 1.93
Expectancy per trade	+0.130R	+0.093R	+0.109R
Profit factor · t-statistic (mean R)	1.34 · 3.10	1.32 · 2.95	1.33 · 4.27
Sharpe ratio (95% CI, block bootstrap)	1.40 (0.61 – 2.14)	1.34 (0.57 – 2.03)	1.53 (0.79 – 2.21)
Sortino ratio	2.80	2.87	3.42
Return per year	+20.8R	+21.5R	+42.3R
Max drawdown · days under water	12.8R · 169	15.2R · 397	26.7R · 195
Calmar (annual R / max DD)	1.63	1.42	1.59
Skew · excess kurtosis (daily)	+1.38 · 2.5	+2.14 · 9.8	+2.05 · 7.2
Positive months	65.5%	67.2%	69.0%
Best · worst month	+11.0R · -8.7R	+14.8R · -6.3R	+20.6R · -13.6R
Expectancy long · short	+0.14R · +0.12R	+0.10R · +0.09R	+0.11R · +0.10R
Correlation: underlying · S&P 500	-0.01 · 0.00	-0.02 · -0.03	-0.01 · -0.02
Correlation with daily move	+0.44	+0.48	+0.52
Deflated - Sharpe p · random-entry p	0.0005 · 0.003	0.0006 · 0.0003	–
Held-back year Sharpe	0.44	1.44	1.14
Second price feed Sharpe	[1.36]	[0.92]	[1.26]

Daily correlation between the two rules: 0.59. Combined by calendar year: 2022 +88R (Sharpe 2.94) · 2023 +58R (2.05) · 2024 +7R (0.27) · 2025 +25R (0.96) · 2026 to Oct +31R (1.21). Profit comes from large moves in either direction (positive skew, no directional or market beta). p-values are deflated for every configuration tried.

Cumulative P&L, Jan 2022 – Oct 2026



Stochastic scaling - 10,000 one-year paths

Risk / trade	Median return	5th pct	Median max DD	95th pct DD	P(DD ≥ 10%)
0.25%	10.5%	0.6%	3.6%	6.7%	0.4%
0.50%	21.0%	1.3%	7.2%	13.5%	18%
0.75%	31.4%	1.9%	10.8%	20.2%	59%
1.00%	41.9%	2.6%	14.4%	27.0%	87%
1.50%	62.9%	3.9%	21.6%	40.4%	99.6%

Stationary block bootstrap (10-day blocks) of the combined daily P&L; risk fixed as a share of starting equity, no compounding. P(losing year) 4.0% at every size [7.4%]. Second feed: median returns ~18% lower.

Prop-firm Monte Carlo - 10,000 simulated 24-month histories, FTMO 2-Step \$200K, live risk settings

61% · 88% funded within 3 · 6 months [58% · 87%]	2.5 mo median time to funded; 52% pass first attempt [2.6 mo · 48%]	\$39.8k median net year 1 · p10 \$7.4k · p90 \$89.7k [32.1k]	\$63.4k median net year 2 · p10 \$17.5k · p90 \$138.5k [49.2k]	4.6% · 0.4% chance of a net loss after 1 · 2 years [6.8% · 0.9%]
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Net of every evaluation fee (3.8 bought over two years on average, re-buys after failures and breaches included), profit splits, payout timing and FTMO's linear scaling plan. Funded accounts are breached about once a year [1.4] and re-bought within the simulation.

MODEL C · RETIRED 5 OCTOBER 2026

Commodity momentum, 5R - the numbers as the original backtest reported them

8 / 8 walk-forward windows profitable (24-month pick, 6-month trade)	PF 1.31 mean out-of-sample profit factor over 2,904 trades	+0.36R expectancy per trade in-sample; +0.23R on 20 held-out months	83% of months profitable, ~40 trades a month, 2022–2026	\$59–128k simulated funded year per \$200k; ~69% challenge pass in ~2 months
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Long-only 5-minute momentum on four commodity CFDs (metals, energy), re-ranked every six months, 5:1 reward-to-risk. Ran live on an FTMO demo account in September–October 2026. What happened next is a separate write-up.